

Abstract

High-performance computing (HPC) systems are essential for scientific research, enabling the processing of large datasets, simulations, and models that would otherwise be infeasible with standard computing resources.

We developed a workflow to automate running machine learning (ML) benchmarks from the MLPerf Inference benchmark suite on HPC systems to quantify the performance of HPC hardware. We sought to make our benchmark workflow reusable, stable, and reproducible. That makes our workflow a valuable tool to HPC because it can assess hardware performance under ML workloads, guide the selection of new hardware, and identify hardware defects.

Introduction

As ML becomes increasingly pertinent in scientific research, HPC systems must be well-optimized for ML tasks. In our project, we implemented an automated workflow to run MLPerf Inference benchmarks^[1] on the Expanse supercomputer at the San Diego Supercomputer Center. Our goal was to measure the throughput of various ML models to quantify the performance of current Expanse hardware under demanding ML tasks.

Materials and Methods

We created our benchmark workflow using the Conda package and environment manager, along with Bash scripts to automate the workflow. That helped ensure that our benchmarks are reproducible. We used the MLCFlow package^[2] to manage the evaluation of the MLPerf Inference benchmarks^[3].

We used Expanse's SLURM workload and job manager to run each of our benchmarks on two different hardware systems:

1. GPU-acceleration with NVIDIA A100-PCIE-40GB
2. GPU-acceleration with Tesla V100-SXM2-32GB

For each benchmark run, MLCFlow collected various metrics: most notably, accuracy and throughput. The model output accuracy had to meet a certain threshold for the run to be considered valid. We recorded the model's throughput, as it is representative of the hardware's performance.

Individual Benchmarks

Bert: Natural Language Processing

- NLP tasks: question answering, sentiment analysis, and language translation

3D-Unet: Medical Image Segmentation

- Medical image segmentation models for kidney tumors
- Based on Kits19 dataset

Stable Diffusion: Text to Image

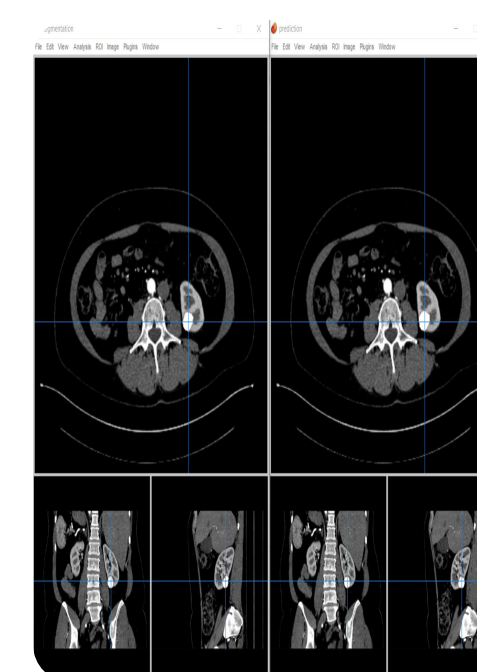
- Model that generates images based off of text descriptions
- Based on the COCO 2014 dataset
- Sample images produced by the model during benchmark runs shown to the right

Mixtral: NLP

- General LLM
- Assess Mixtral's ability to handle diverse NLP workloads with high accuracy and scalability

Whisper: Speech to Text

- OpenAI benchmark to evaluate performance of audio transcription
- Different accents, background noise, speech speeds.



3D-Unet sample data



Stable diffusion: "Two large trains at a modernized train station"

Results

Throughput (samples/sec)

Bert

A100 (mini): 39.2403
V100 (mini): 33.4748

A100 (full): 41.3058
V100 (full): 34.4918

3D-Unet

A100: 0.287
V100: 0.126

Stable Diffusion

A100: 0.370
V100: 0.172

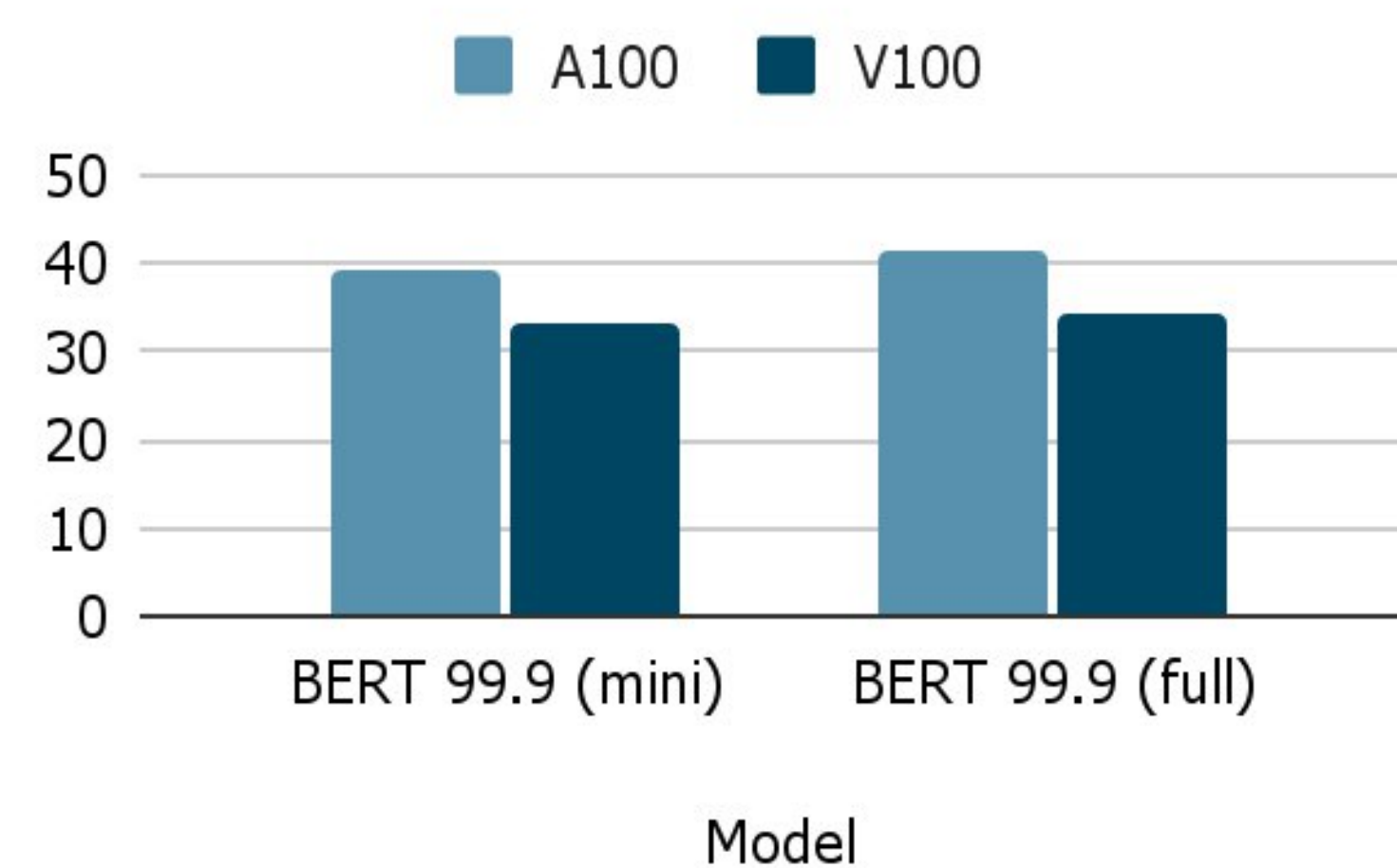
Mixtral

Benchmarking in progress due to implementation complexity.

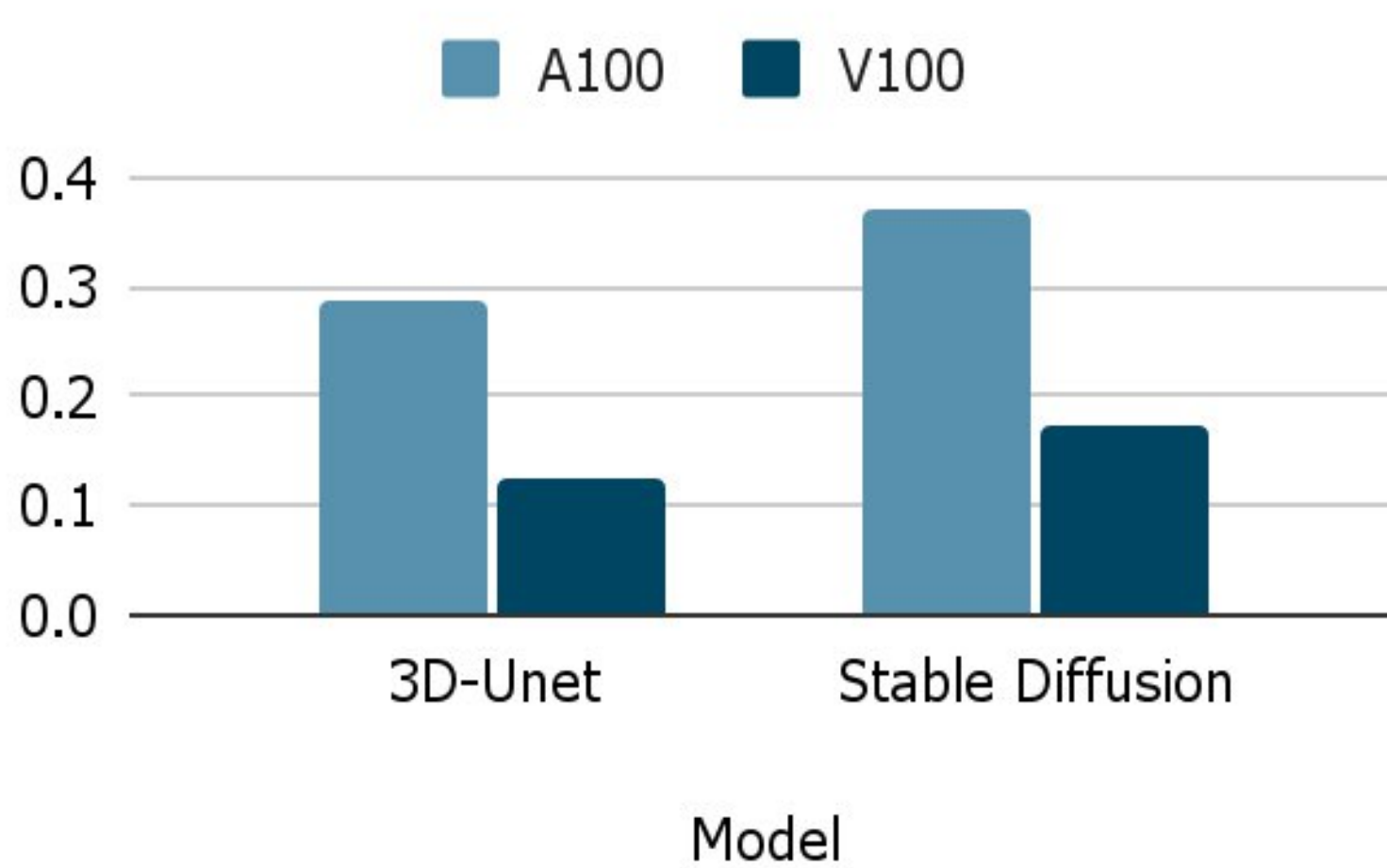
Whisper

Benchmarking in progress due to being a novel model.

Throughput (samples/sec)



Throughput (samples/sec)



References

- [1] MLCommons, "MLPerf Inference Benchmarks," <https://docs.mlcommons.org/inference/>
- [2] MLCommons, "MLCFlow Documentation," <https://docs.mlcommons.org/mlcflow/>
- [3] MLCommons, "Reference Implementations of MLPerf Inference Benchmarks,"
- [4] MLCommons, "MLCommons, Inference Datacenter," <https://public.tableau.com/app/profile/data.visualization6666/viz/MLCommons-InferenceDatacenter/MLCommons-Inference>

Conclusion

All benchmark runs show the A100s outperform the V100s by a margin that aligns with published data from NVIDIA. However, the throughput is noticeably slower than other published results^[4], which is likely attributed to a less optimized inference implementation.

Applications & Future Work

In the future, we intend to also produce results for CPU inference to compare to the A100s and V100s. We also intend to publish our benchmark results in the future to MLCommons to serve as a reference point for other researchers with similar hardware to the Expanse supercomputer.